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MAT 210: Discrete Mathematics

Institution: Harborview Community College (fictional) **Credits:** 3 (semester hours) **Instructor:** Staff
Prerequisite: MAT 140 or placement **Contact hours:** 3 lecture hours per week, 15 weeks

Course description

Discrete mathematics for computer science majors (3 credits): logic, sets, proof techniques including induction, functions and relations, basic counting (permutations and combinations) and an introduction to graphs and trees. Recurrence relations and discrete probability are not covered.

Learning outcomes

- Use propositional and predicate logic to express statements
- Write direct, contrapositive, contradiction and induction proofs
- Work with sets, functions and relations
- Solve counting problems with permutations, combinations and the pigeonhole principle
- Describe basic properties of graphs and trees

Weekly schedule

Week	Topic
1	Propositional logic and truth tables
2	Predicates and quantifiers
3	Direct proof and proof by contrapositive
4	Proof by contradiction; proof writing workshop
5	Mathematical induction
6	Strong induction; exam 1
7	Sets and set operations
8	Functions: injective, surjective, bijective
9	Relations; equivalence relations and partial orders
10	Counting: sum and product rules
11	Permutations and combinations
12	Pigeonhole principle; exam 2
13	Introduction to graphs: degree, paths, connectivity
14	Trees and spanning trees
15	Review; final exam

Assessment

Homework 30%, two exams 40%, final exam 30%. Note: recurrence relations and discrete probability are not part of this course.

Textbook: instructor-provided open course notes.

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